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A Drug-Environment Interaction Between PM_{2.5} Concentration and Corticosteroid Use on Cardiovascular and Thromboembolic Events in Older Adults

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ABSTRACT

Background: Few studies have investigated the interaction between drug-environment despite the plausibility and concern. We sought to explore the synergistic effects of corticosteroids and PM_{2.5} on cardiovascular and thromboembolic events in high-risk older adults in Taiwan.

Objectives: To assess the synergistic effect of corticosteroid use and PM_{2.5} exposure on cardiovascular and thromboembolic events and all cause mortality among elderly individuals in Taiwan.

Methods: We conducted a retrospective cohort study using the Taiwan National Health Insurance database from 2009 to 2019. We included patients aged 65 years or older and at high risk for cardiovascular and thromboembolic events. Exposures included corticosteroid therapy and seasonal mean PM_{2.5}. Primary outcomes included myocardial infarction or acute coronary syndrome, ischemic stroke or transient ischemic attack, heart failure, venous thromboembolism, atrial fibrillation, and all-cause mortality. We then fitted history-adjusted marginal structural Cox proportional hazard models to investigate both the independent and synergistic effects of PM_{2.5} and corticosteroid use on the CTE outcomes. We measured whether there was a sufficient cause interaction between PM_{2.5} and corticosteroids using estimates of relative excess risk due to interaction (RERI).

Results: The cohort included 373 402 participants, with average age at index date 73.6 ± 7.2 years; 57.7% of cohort members were female. During the study period, 1 614 331 (43.2%) participants had at least one corticosteroid prescription. The average PM_{2.5} concentration was 28.3 ± 17.4 µg/m³. Increasing PM_{2.5} from 15 to 20 µg/m³ resulted in significantly increased excess risk due to interaction (RERI [95% CI]) for all-cause mortality (3.1% [0.27%, 5.9%]).

Conclusions: A synergistic effect between seasonal PM_{2.5} exposure and corticosteroid use was found on all-cause mortality in elderly patients at high risk for cardiovascular and thromboembolic events.

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Highlights

- Few studies have investigated the interaction between drug-environment factors despite the plausibility and concern.
- Considering that a majority of elderly individuals with pulmonary disease are prescribed corticosteroids, it is imperative to examine the potential interaction between air pollution exposure and steroid use in this population.
- The study's database covered a representative 99% of the national population health data and was followed for 10 years to assess whether there is a synergistic effect of long-term exposure to PM_{2.5} and steroid use on increasing the risk of cardiovascular and thromboembolic effects and death.
- Our study suggests that the concurrent exposures of steroids and PM_{2.5} may have synergistic effects that increase the risk of cardiovascular and thromboembolic events as well as all-cause mortality in older adults.
- For these vulnerable older adults, stricter control of PM_{2.5} exposure is essential to reduce health risks. Additionally, given the potential synergistic effects between corticosteroids and PM_{2.5}, cautious prescribing should not be overlooked, and the option to de-prescribe corticosteroids could also be considered in older adults.

Plain Language Summary

Limited research has been conducted on the interaction between drugs and environmental factors. Our study aimed to investigate the combined effects of corticosteroids and PM_{2.5} on cardiovascular and thromboembolic incidents among high-risk older adults in Taiwan. We performed a study utilizing the National database from 2009 to 2019, including patients aged 65 and older. The exposures of interest were corticosteroid treatment and seasonal average PM_{2.5} levels. The primary outcomes measured were cardiovascular and thromboembolic events and overall mortality. The cohort comprised 373 402 participants, with an average age of 73.6 ± 7.2 years, and 57.7% were female. Throughout the study, 161 433 (43.2%) participants received at least one corticosteroid prescription. The average PM_{2.5} concentration was 28.3 ± 17.4 µg/m³. An increase in PM_{2.5} from 15 to 20 µg/m³ was associated with a significant rise in risk for overall mortality. Our findings indicate a synergistic effect between seasonal PM_{2.5} exposure and corticosteroid use on overall mortality in elderly patients at high risk for cardiovascular and thromboembolic events.

1 | Introduction

Few studies have investigated the interaction between drug-environment despite the plausibility and concern [1]. In recent decades, increased fuel and transport use due to industrialization and urbanization has led to an increase in air pollutant

emissions throughout Asia [2]. Many studies have demonstrated the health impacts of air pollution [3, 4]. Air pollution exerts a disproportionate impact on the elderly population, exacerbating or precipitating respiratory conditions such as asthma and chronic obstructive pulmonary disease (COPD). Additionally, it elevates the risk of cardiovascular incidents, including myocardial infarctions and cerebrovascular accidents, and is implicated in the progression of cognitive decline, dementia, and depressive disorders. The effects of air pollution, particularly PM_{2.5}, on the cardiovascular system are consistently reported; PM_{2.5} exposure is associated with an increased risk of cardiovascular and thromboembolic events such as stroke, myocardial infarction, heart failure, venous thromboembolism, and other cardiovascular events [5–17].

In Taiwan, cardiovascular disease is the second leading cause of death, and older adults with high-risk medical conditions constitute a vulnerable population requiring special attention. Corticosteroids used to treat COPD and asthma are known to cause cardiovascular and thromboembolic events. The effect of corticosteroids on cardiovascular events has been demonstrated in several studies, with the primary mechanism being the effect of corticosteroids on blood vessels [18–21]. Evidence also suggests a causal relationship between corticosteroid exposure and thromboembolic events by increasing clotting factors and fibrinogen levels, including pulmonary embolism and venous thromboembolism [22].

However, studies investigating the interaction between corticosteroids and PM_{2.5} exposure on the risk of cardiovascular and thromboembolic events and mortality in older adults are scarce. One recent study evaluated the synergistic effects of corticosteroids and PM_{2.5} on cardiovascular and thromboembolic events in high-risk older adults using Medicare databases in the US. The findings suggest that the effects of PM_{2.5} and corticosteroid exposure on heart failure and myocardial infarction/acute coronary syndrome are synergistic [23]. The average PM_{2.5} level during the study period was 8.3 ± 2.4 µg/m³, which is relatively low compared to East and Southeast Asia, where a large portion of the world's population is concentrated and PM_{2.5} levels are routinely greater than 100 µg/m³ [24]. In the present study, the synergistic effects of corticosteroids and PM_{2.5} on cardiovascular and thromboembolic events will be evaluated using the Taiwan National Health Insurance Database (NHID).

2 | Materials and Methods

2.1 | Study Population and Cohort Definition

Our study used data from the Taiwan NHID covering the period from 2009 to 2019. The NHID is a comprehensive health insurance program that provides coverage for 99.9% of Taiwan's population. The database includes beneficiary registries, ambulatory care claims, inpatient claims, and prescription dispensing claims from pharmacies. The study cohort consisted of patients with chronic conditions who are at high risk for cardiovascular and thromboembolic events, including patients with atrial fibrillation (AF), a history of heart failure (HF), previous carotid stenosis, coronary artery disease

(CAD), cancer, peripheral vascular disease (PVD), previous cerebrovascular accidents (CVA), previous venous thromboembolisms (VTE), and those who had total joint arthroplasty. The index date was defined as the date when the patient first established an outpatient or inpatient medical record between 2009 and 2018. If any of the aforementioned chronic disease diagnoses appeared in inpatient records within 12 months prior to the index date, the patient was included in the study cohort. A detailed definition of the high-risk cohort is available in Supplement Table 1 of the Supplementary Appendix. The study was approved by the National Cheng Kung University Hospital Institutional Review Board (IRB no: B-EX-108-056).

2.2 | PM_{2.5} Exposures

Seasonal average PM_{2.5} concentrations from the 2009–2018 Data Bank for Atmospheric and Hydrologic Research were linked to the NHID based on postcode. The seasons of Winter, Spring, Summer, and Fall were defined as occurring from December to February, March to May, June to August, and September to November, respectively. Nationwide estimates of seasonal average PM_{2.5} concentrations were calculated based on hourly data collected at 85 monitoring stations. Hourly PM_{2.5} concentrations were then averaged within each combination of postcode and season to obtain postcode-level seasonal average concentrations. For areas with no monitoring station, the PM_{2.5} concentration at the nearest monitoring station was used.

To identify each patient's postcode of residence, we employed a residence estimation algorithm developed by Lin et al. [25], which had been validated against census data (Pearson correlation coefficient 0.960). The rule of residence estimation was based on the classification of beneficiaries in the NHI program and the location of hospital or clinic visits. This estimation was composed of four steps: (1) if the employment status of a beneficiary belongs to a certification requiring a professional license (such as accountant, lawyer, health care personnel), farmers, fishermen, veterans, or low-income citizens, whose residence can be designated as identical to the one registered in the NHI registry, as professional practice or veteran, and social benefit is restricted to the location of a city or county. The residence estimation algorithm is shown in Supplement Figure 1 of the Supplementary Appendix. (2) for those who did not belong to the above categories, we retrieved the ambulatory care for upper respiratory tract infection (URI) in NHID, and assigned the location of hospital/clinic they visited as their residence area; (3) if the person had no records for ambulatory URI, we checked for other ambulatory visits in the hospital/clinic and selected the one with the highest number of visits as their residence location; (4) for those who had no claims record of ambulatory care, the residence information in the most recent registry was considered to be their residence location.

2.3 | Corticosteroid Exposures

Systemic corticosteroids of interest included cortisone, hydrocortisone, prednisone, prednisolone, methylprednisolone,

triamcinolone, dexamethasone, and betamethasone. We identified systemic corticosteroid exposure by using ATC code in prescription dispensing claims from pharmacies. Each person's baseline was 12 months after their enrollment during 2009 to 2018. We identified the first corticosteroid use after the index date. Since the majority of corticosteroid use is intermittent for COPD, asthma, or flare of autoimmune diseases and the risk of adverse effects is not considered to diminish with repeated use in theory [26], we did not require new use. To accommodate computational impracticality of daily level data, the index corticosteroid therapy was identified and then updated on a quarterly basis until each participant reached one of the study endpoints. To ensure that individuals' quarterly follow-up times aligned with initiation dates of corticosteroid usage, we developed individualized drug exposure panels for each cohort member. For corticosteroid users, the initiation of these panels corresponded with the commencement of corticosteroid therapy during the follow-up period, whereas for non-users, the panels began at seasonal transitions. Specifically, for participants who received corticosteroids at any point during follow-up, the first day of their corticosteroid treatment was aligned with the first day of a quarter. Subsequently, we generated participant-specific panels covering quarterly intervals, extending retrospectively to the participant's index date and prospectively to their end date (Supplement Figure 2 of the Supplementary Appendix).

2.4 | Outcomes

We followed all participants until they developed one of the outcomes of interest, or until they experienced a censoring event. Outcome of interest included cardiovascular and thromboembolic events outcomes and all-cause mortality. Cardiovascular and thromboembolic events were defined as events with at least one relevant inpatient diagnosis identified by an ICD-9/10 code in the first or second position. Cardiovascular and thromboembolic events included acute coronary syndrome (including myocardial infarction) (ACS/MI), ischemic stroke/transient ischemic attack (IS/TIA), atrial fibrillation (AF), deep vein thrombosis (DVT), and heart failure (HF). Specific criteria and details of the CTE definitions are provided in Supplement Table 2 of the Supplementary Appendix. Censoring occurred due to end of the study period, or after the discontinuation of their first corticosteroid therapy, with a 14-day grace period.

2.5 | Covariates

Baseline variables at the individual level including age, sex, and insurance premium (a proxy for socioeconomic status) were obtained from the NHID. Information on the presence or absence of baseline comorbidities and health behaviors was obtained using outpatient and inpatient diagnosis codes from the NHID. These variables were considered as time-invariant confounders (see Supplement Table 3 in the Supplementary Appendix). Measures of healthcare utilization within the 12 months preceding the index date. Including the number of hospitalizations, total hospitalization days, number of emergency department visits, number of outpatient visits, and number of medications

dispensed were also taken from the NHID. Season and year were treated as time-varying confounders.

2.6 | Statistical Analysis

We began by conducting descriptive analyses of individual characteristics and prevalence of corticosteroid use. We calculated the number of person-years at risk, number of events, and event rates per 1000 person-years for each of the six outcomes examined, both overall and stratified by corticosteroid status. We then fitted history-adjusted marginal structural Cox proportional hazard models, facilitated by fitted inverse probability weights (IPWs), to investigate both the independent and synergistic effects of PM_{2.5} and corticosteroid use on the CTE outcomes. We fitted history-adjusted marginal structural Cox models (HAMSMs), which use inverse probability weights (IPWs) to adjust for time-varying confounding [23]. Unlike standard marginal structural models that account for the full cumulative exposure history, HAMSMs estimate the association between each outcome and the exposure level at a given time point. This approach is particularly well-suited to the present analysis, which involves two simultaneous time-varying exposures, PM_{2.5} and corticosteroid use, as well as their interaction; time-varying covariates may confound one exposure while

being influenced by the other, rendering standard regression adjustment inadequate. Details of the IPW estimation procedure are provided in the Supplementary Appendix. We assessed synergy between PM_{2.5} and corticosteroids by calculating the relative excess risk due to interaction (RERI), a measure of interaction on the additive scale that can be interpreted as the relative increase in the combined effect of the two exposures versus the individual effects of the two exposures summed together (presented as a percentage). The delta method and robust variance estimates of the Cox model parameters were used to compute standard errors for the RERI. Additional details for estimating the RERI are provided in the Supplementary Appendix. PM_{2.5} 15 µg/m³ is the policy-relevant threshold in Taiwan, while 23 µg/m³ represents Taiwan's median value. We selected 15 and 23 µg/m³ to determine how much the risk of cardiovascular disease increases when PM_{2.5} concentrations rise from the regulated level of 15 µg/m³ to the current level of 23 µg/m³. We employed regression models to calculate the hazard ratios for PM_{2.5} and corticosteroids across various outcomes.

3 | Results

The cohort included 373 402 participants, with average age at index date 73.6 ± 7.2 years; 57.7% of cohort members were female.

TABLE 1 | Characteristics of high CTE risk cohort.

	All participants (n = 373 402)	Participants who received corticosteroids (n = 161 431)	Participants who never received corticosteroids (n = 211 971)
Age at index date	73.6 ± 7.2	73.1 ± 6.8	74 ± 7.5
Male	157 948 (42.3)	67 906 (42.1)	90 042 (42.5)
Chronic conditions			
Total joint arthroplasty	181 606 (48.6)	85 361 (52.9)	96 245 (45.4)
Acute coronary syndrome	10 772 (2.9)	4 453 (2.8)	6 319 (3.0)
Myocardial Infarction	10 551 (2.8)	3 591 (2.2)	6 960 (3.3)
Cancer	91 450 (24.5)	339 353 (24.4)	52 097 (24.6)
Atrial Fibrillation	26 626 (7.1)	9 302 (5.8)	17 324 (8.2)
All Stroke	44 732 (12.0)	15 966 (9.9)	28 766 (13.6)
Ischemic Stroke	39 381 (10.5)	14 072 (8.7)	25 309 (11.9)
Transient Ischemic Attack	10 773 (2.9)	4 616 (2.9)	6 157 (2.9)
Heart Failure	41 472 (11.1)	13 919 (8.6)	27 553 (13.0)
Peripheral Vascular Disease	1 214 (0.3)	486 (0.3)	728 (0.3)
Venous Thromboembolism	4 698 (1.3)	2 007 (1.2)	2 691 (1.3)
Carotid Stenosis	3 713 (1.0)	1 443 (0.9)	2 270 (1.1)
Health Service Measures			
Prior Inpatient Hospitalizations	1.7 ± 1.6	1.7 ± 1.5	1.8 ± 1.7
Prior ER Visits	1 ± 1.7	0.9 ± 1.6	1.1 ± 1.7
Prior Outpatient Visits	36.1 ± 21.8	39.6 ± 22.8	33.4 ± 20.7
Prior Medications Dispensed	13.6 ± 5.8	13.8 ± 5.6	13.4 ± 5.9

Note: Numbers are count (%) for categorical variables or mean ± standard deviation for continuous variables.

During the study period, 1 614 331 (43.2%) participants had at least one corticosteroid prescription. The median duration of corticosteroid use was 7 days, with a 75th percentile of 28 days and a 90th percentile of 80 days. Individual characteristics and CTE risk conditions were similar between corticosteroid users

and nonusers (Table 1). Corticosteroid users had more outpatient visits than non-users (39.6 ± 22.8 vs. 33.4 ± 20.7). Supplement Table 4 of the Supplementary Appendix shows summary statistics of comorbidities included in the random forest IPW models. The average $PM_{2.5}$ concentration was $28.3 \pm 17.4 \mu\text{g}/\text{m}^3$ (median

TABLE 2 | Incidence rates of cardiovascular thromboembolic events in overall cohort and by systemic steroid use.

Outcome	All		Participants who received corticosteroids		Participants who never received corticosteroids	
	Person-Years at Risk	Events (Per 1000 Person-Years)	Person-Years at Risk	Events (Per 1000 Person-Years)	Person-Years at Risk	Events (Per 1000 Person-Years)
MI or ACS	1 169 272	10 376 (8.9)	13 321	241 (18.1)	1 155 951	10 135 (8.8)
Ischemic Stroke or TIA	1 134 862	22 314 (19.7)	12 971	374 (28.8)	1 121 891	21 940 (19.6)
Heart Failure	1 149 847	22 850 (19.9)	12 908	789 (61.1)	1 136 939	22 061 (19.4)
Venous Thromboembolism	1 187 503	2465 (2.1)	13 499	104 (7.7)	1 174 004	2361 (2.0)
Atrial Fibrillation	1 172 901	8439 (7.2)	13 326	248 (18.6)	1 159 574	8191 (7.1)
Death	1 192 927	87 834 (73.6)	13 578	2796 (205.9)	1 179 349	85 038 (72.1)

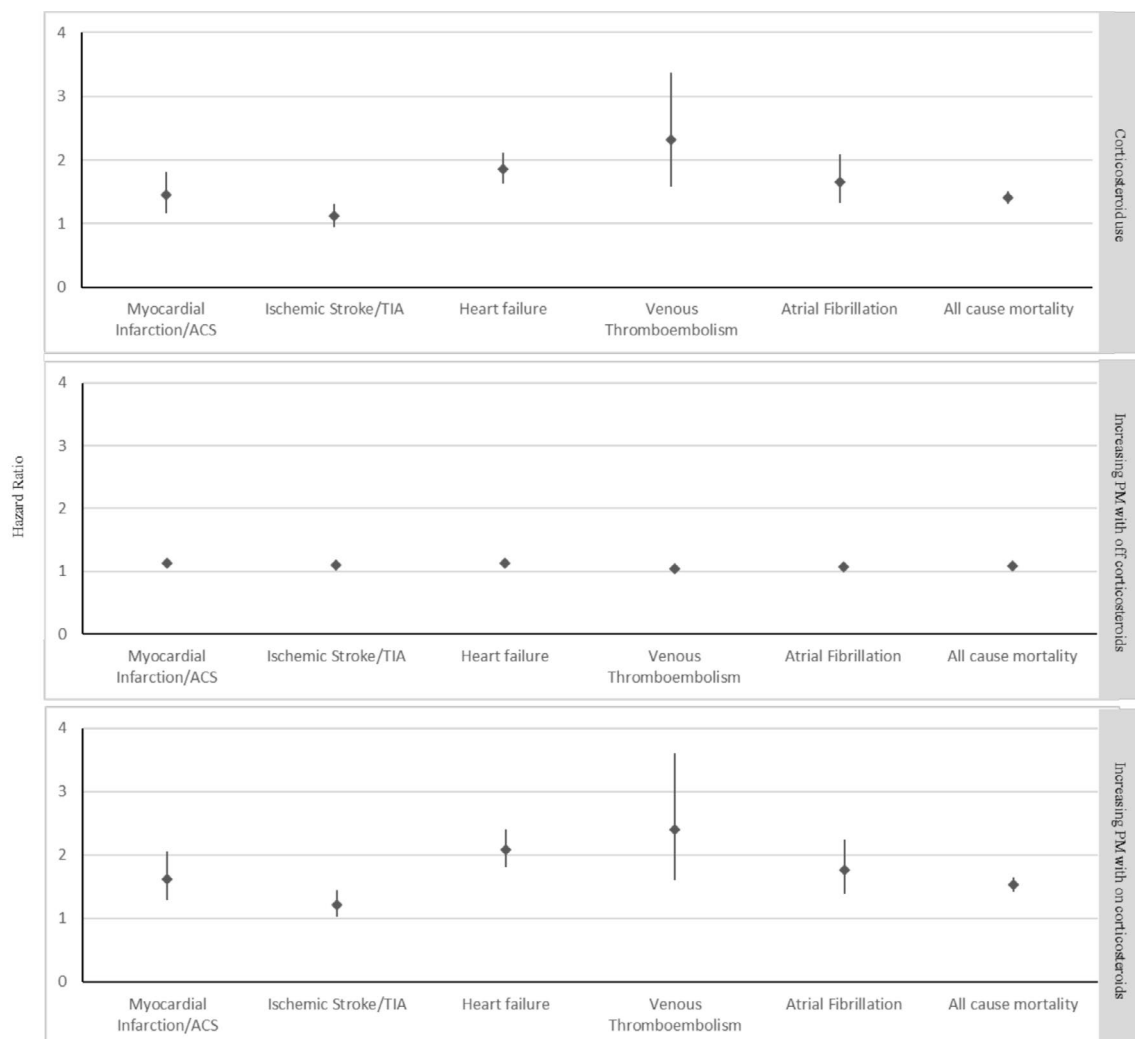


FIGURE 1 | Hazard ratios for corticosteroid use and increasing $PM_{2.5}$ (15 vs. $23 \mu\text{g}/\text{m}^3$) while on and off corticosteroids.

concentration $23.5\mu\text{g}/\text{m}^3$). $\text{PM}_{2.5}$ concentrations were highest in winter and spring ($30.8\pm 6.2\mu\text{g}/\text{m}^3$ and $27.2\pm 5.9\mu\text{g}/\text{m}^3$, respectively); the levels in summer and fall were $14.1\pm 4.1\mu\text{g}/\text{m}^3$ and $24.4\pm 5.1\mu\text{g}/\text{m}^3$, respectively. We used a restricted cubic spline curve, which showed a plateau curve around $\text{PM}_{2.5}$ levels of $15\text{--}40\mu\text{g}/\text{m}^3$ (see Supplement Figure 3 of the Supplementary Appendix).

During the follow up period, we observed a total of 10376 for MI/ACS, 22314 for ischemic stroke/TIA, 22850 for HF, 2465 for VTE, 8439 for AF and 87834 for death. Incidence rates of all outcomes were higher in corticosteroid users than non-users. Incidence rates per 1000 person-years in corticosteroid users and non-users respectively were 18.1 versus 8.8 for MI/ACS, 28.8 versus 19.6 for ischemic stroke/TIA, 61.1 versus 19.4 for HF, 7.7 versus 2.0 for VTE, 18.6 versus 7.1 for AF, and 205.9 versus 72.1 for death (Table 2).

When $\text{PM}_{2.5}$ was fixed at $23\mu\text{g}/\text{m}^3$ (close to the median $\text{PM}_{2.5}$ in Taiwan), the hazard ratios (95% CI) for corticosteroid use were 1.45 (1.16, 1.82) for MI/ACS, 1.112 (0.94, 1.31) for ischemic stroke/TIA, 1.85 (1.62, 2.12) for heart failure, 2.31 (1.59, 3.37) for VT, 1.66 (1.32, 2.08) for AF and 1.40 (1.30, 1.51) for all-cause mortality. Increasing the $\text{PM}_{2.5}$ concentration from 15 to $23\mu\text{g}/\text{m}^3$ in the absence of corticosteroid exposure resulted in hazard ratios of 1.12 (1.11, 1.13) for MI/ACS, 1.09 (1.08, 1.10) for ischemic stroke/TIA, 1.12 (1.11, 1.13) for heart failure, 1.04 (1.01, 1.07) for VT, 1.06 (1.05, 1.08) for AF and 1.09 (1.08, 1.09) for all-cause mortality. When considering $\text{PM}_{2.5}$ increases along with corticosteroid use, the risk of CTE and all-cause mortality was higher than when considering $\text{PM}_{2.5}$ and corticosteroid use alone (Figure 1). Increasing $\text{PM}_{2.5}$ from 15 to $20\mu\text{g}/\text{m}^3$ resulted in significantly increased excess risk due to interaction (RERI [95% CI]) for all-cause mortality (3.1% [0.3%, 5.9%]) (Figure 2).

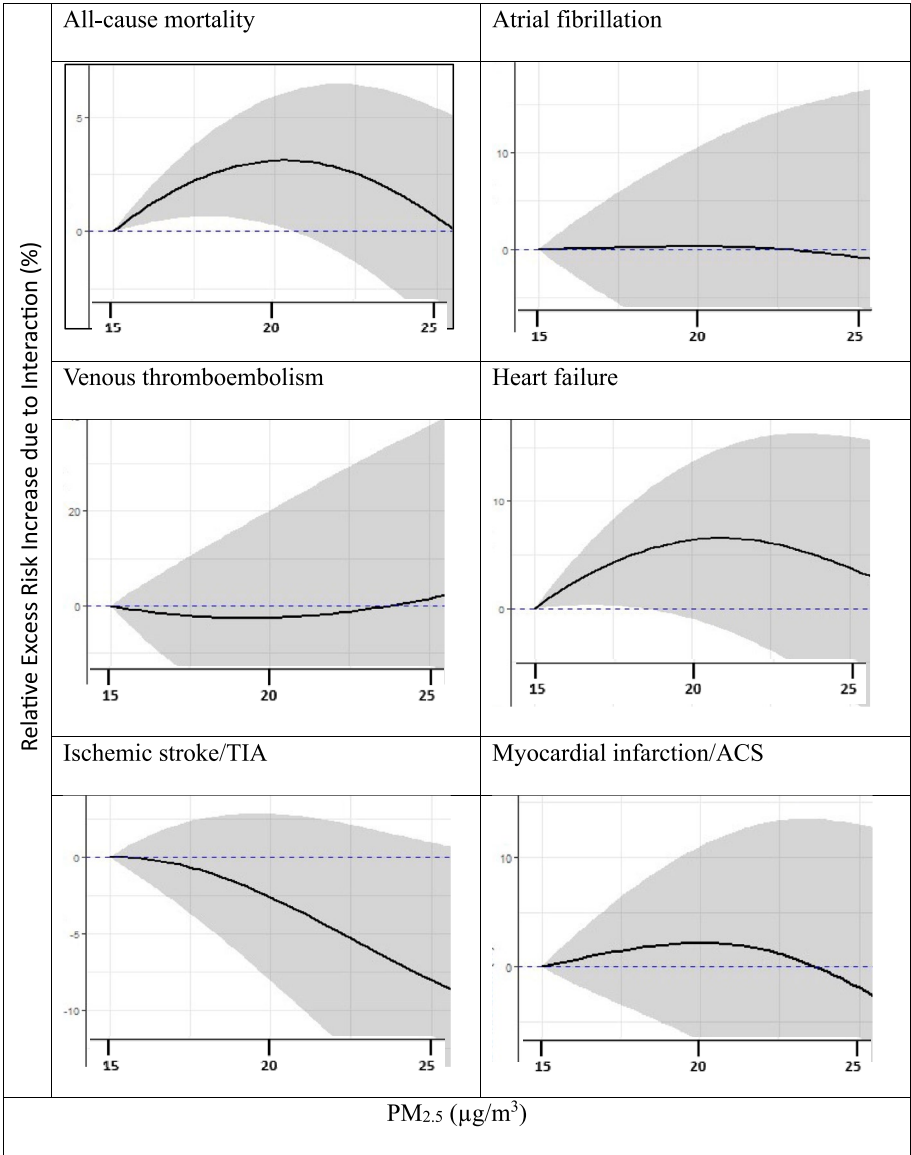


FIGURE 2 | Relative excess risk due to interaction (RERI) between $\text{PM}_{2.5}$ exposure and corticosteroid use for each of the six outcomes (and their 95% confidence interval bands) with $\text{PM}_{2.5}$ concentrations to reference values of $15\mu\text{g}/\text{m}^3$. Source: Curves represent the change in RERI due to simultaneously initiating corticosteroid treatment and increasing $\text{PM}_{2.5}$ exposure to any given level above the corresponding $\text{PM}_{2.5}$ reference level.

4 | Discussion

In a retrospective study using administrative health and air pollution data over an 11-year period in Taiwan, we found a synergistic effect of concurrent exposure to PM_{2.5} and corticosteroid use on all-cause mortality, which is consistent with the US study by Josey et al. (12.5% [−0.6%, 25.5%]) [23]. Risk escalation on all-cause mortality during concurrent high exposure of PM_{2.5} and corticosteroid use was larger than what had been estimated from the independent effects of the two exposures summed together.

PM_{2.5} can be distributed directly through the bloodstream to distant target organs, causing localized oxidative stress, inflammation, and atherosclerotic plaques, eventually leading to thrombosis [27]. Steroids may stimulate fibrinogen activator inhibitor-1 and reduce fibrinolysis [28]. Both mechanisms induce hypercoagulable states, so concomitant exposure may produce synergistic effects. Our results did not show a significant interaction effect of PM_{2.5} exposure and corticosteroid use on cardiovascular and thromboembolic events outcomes, but we found a trend toward an interaction effect of these two factors on heart failure and myocardial infarction/ACS, which is consistent with previous results from a US Medicare population [23]. Among them, significant interactions were detected with increasing PM_{2.5} from 8 to 12 µg/m³ for HF (RERI 15.6% [95% CI 4%, 27.3%]) [23].

Our analyses showed that exposures to high concentrations of PM_{2.5} do not increase cardiovascular and thromboembolic events risk as expected. This inconsistent finding could be attributed to the differences in baseline characteristics between our cohort and the US cohort. Although the inclusion criteria for the two studies were similar, patients included in the present study were slightly younger (73.6 vs. 76.8) and had substantially lower rates of AF (7.1% vs. 30.8%), HF (11.1% vs. 15.1%), myocardial infarction (2.8% vs. 6.5%), and peripheral vascular disease (0.3% vs. 10.1%) at baseline. In addition, there is a non-linear relationship between PM_{2.5} concentration and cardiovascular and thromboembolic events [29, 30]. Based on the results of the cubic spline curve analysis, our findings could be explained by a nonlinear relationship between PM_{2.5} exposure and cardiovascular and thromboembolic events.

Another issue that may have led to inconsistencies between our findings and those from the US relates to prescribing patterns for corticosteroids. A study by Wallace et al. on oral corticosteroid prevalence and prescribing pattern in the US and Taiwan in 2009–2018 found that the median (IQR) annual dose and duration were much higher in the US (200.0 [105.0–310.0] mg and 6.0 [5.0–12.0] days) than in Taiwan (58.2 [37.5–97.5] mg and 5.0 [3.0–9.0] days) [31]. Their report suggested that the levels of corticosteroid exposures in our cohort were likely lower than in the US studies, which might have contributed to the inconsistent results.

Although we were able to address multiple potential sources of bias, several limitations to our study should be noted. First, exposure data were obtained from fixed monitoring stations, and patient residence was indirectly estimated from claims data, both of which introduce the potential for exposure misclassification. To mitigate this bias, we used the most frequently visited clinic as a surrogate for patient residence. Additionally, the lack of socioeconomic data in the NHID limited our ability to

account for individual-level measures taken to mitigate air pollution exposure, such as reducing outdoor exposure during periods of poor air quality. In addition, the survey of Family Income and Expenditure in Taiwan reports that the proportion of households with air purifiers increased from 10.9% in 2009 to 26.4% in 2022 [32], which could have introduced bias in exposure measurement, as it would have affected individual exposure to PM_{2.5} in a time-varying manner. Although we were unable to link the survey results to NHID, we were able to address this limitation to some extent by using insurance premium (which reflects income) as a proxy for health behavior.

5 | Conclusions

A synergistic effect between seasonal PM_{2.5} exposure and corticosteroid use was found on all-cause mortality in elderly patients at high risk for cardiovascular and thromboembolic events.

Author Contributions

Conceptualization (Soko Setoguchi, David Robinson), data curation (Shu-Ping Huang, Chien-Chou Su), formal analysis (Shu-Ping Huang, Chien-Chou Su), funding acquisition (Soko Setoguchi), investigation (David Robinson, Rachel Nethery, Soko Setoguchi), methodology (Rachel Nethery, Kevin Josey, Soko Setoguchi, Chien-Chou Su), project administration (Soko Setoguchi, Melanie Rua), resources (Yea-Huei Kao Yang), software (Chien-Chou Su), supervision (David Robinson, Soko Setoguchi, Rachel Nethery, Benjamin Bates, Yea-Huei Kao Yang), validation (Soko Setoguchi, Benjamin Bates, Yea-Huei Kao Yang), visualization (Shu-Ping Huang, Chien-Chou Su), writing – original draft (Shu-Ping Huang, Chien-Chou Su), writing review and editing (all authors).

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Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from Ministry of Health and Welfare. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the author(s) with the permission of Ministry of Health and Welfare.

References

- Centers for Disease Control and Prevention, “Heat and Medications – Guidance for Clinicians,” <https://www.cdc.gov/heat-health/hcp/clinical-guidance/heat-and-medications-guidance-for-clinicians.html>.

2. Economic and Social Commission for Asia and the Pacific, "Air Quality in Asia: Air Pollution Trends and Mitigation Policy Options," <https://www.unescap.org/kp/2023/air-quality-asia-air-pollution-trends-and-mitigation-policy-options>.
3. World Health Organization, "COP26 Special Report on Climate Change and Health: The Health Argument for Climate Action," <https://www.who.int/publications/i/item/9789240036727>.
4. I. Manisalidis, E. Stavropoulou, A. Stavropoulos, and E. Bezirtzoglou, "Environmental and Health Impacts of Air Pollution: A Review," *Frontiers in Public Health* 8 (2020): 14, <https://doi.org/10.3389/fpubh.2020.00014>.
5. S. Park, R. Allen, and C.-H. Lim, "A Likely Increase in Fine Particulate Matter and Premature Mortality Under Future Climate Change," *Air Quality, Atmosphere & Health* 13 (2020): 143–153, <https://doi.org/10.1007/s11869-019-00785-7>.
6. Q. Di, L. Dai, Y. Wang, et al., "Association of Short-Term Exposure to Air Pollution With Mortality in Older Adults," *Journal of the American Medical Association* 318, no. 24 (2017): 2446–2456, <https://doi.org/10.1001/jama.2017.17923>.
7. Q. Di, Y. Wang, A. Zanobetti, et al., "Air Pollution and Mortality in the Medicare Population," *New England Journal of Medicine* 376, no. 26 (2017): 2513–2522, <https://doi.org/10.1056/NEJMoa1702747>.
8. B. Wang, K. D. Eum, F. Kazemiparkouhi, et al., "The Impact of Long-Term PM(2.5) Exposure on Specific Causes of Death: Exposure-Response Curves and Effect Modification Among 53 Million U.S. Medicare Beneficiaries," *Environmental Health* 19, no. 1 (2020): 20, <https://doi.org/10.1186/s12940-020-00575-0>.
9. Y. Wei, M. D. Yazdi, Q. Di, et al., "Emulating Causal Dose-Response Relations Between Air Pollutants and Mortality in the Medicare Population," *Environmental Health* 20, no. 1 (2021): 53, <https://doi.org/10.1186/s12940-021-00742-x>.
10. X. Wu, D. Braun, J. Schwartz, M. A. Kioumourtzoglou, and F. Dominici, "Evaluating the Impact of Long-Term Exposure to Fine Particulate Matter on Mortality Among the Elderly," *Science Advances* 6, no. 29 (2020): eaba5692, <https://doi.org/10.1126/sciadv.aba5692>.
11. J. Chen and G. Hoek, "Long-Term Exposure to PM and All-Cause and Cause-Specific Mortality: A Systematic Review and Meta-Analysis," *Environment International* 143 (2020): 105974, <https://doi.org/10.1016/j.envint.2020.105974>.
12. C. A. Pope, 3rd, R. T. Burnett, M. C. Turner, et al., "Lung Cancer and Cardiovascular Disease Mortality Associated With Ambient Air Pollution and Cigarette Smoke: Shape of the Exposure-Response Relationships," *Environmental Health Perspectives* 119, no. 11 (2011): 1616–1621, <https://doi.org/10.1289/ehp.1103639>.
13. F. Dominici, R. D. Peng, M. L. Bell, et al., "Fine Particulate Air Pollution and Hospital Admission for Cardiovascular and Respiratory Diseases," *JAMA* 295, no. 10 (2006): 1127–1134, <https://doi.org/10.1001/jama.295.10.1127>.
14. J. Rhee, F. Dominici, A. Zanobetti, et al., "Impact of Long-Term Exposures to Ambient PM(2.5) and Ozone on ARDS Risk for Older Adults in the United States," *Chest* 156, no. 1 (2019): 71–79, <https://doi.org/10.1016/j.chest.2019.03.017>.
15. M. Yitshak-Sade, J. F. Bobb, J. D. Schwartz, I. Kloog, and A. Zanobetti, "The Association Between Short and Long-Term Exposure to PM(2.5) and Temperature and Hospital Admissions in New England and the Synergistic Effect of the Short-Term Exposures," *Science Total Environment* 639 (2018): 868–875, <https://doi.org/10.1016/j.scitotenv.2018.05.181>.
16. S. S. Joshi, M. R. Miller, and D. E. Newby, "Air Pollution and Cardiovascular Disease: The Paul Wood Lecture, British Cardiovascular Society 2021," *Heart* 108, no. 16 (2022): 1267–1273, <https://doi.org/10.1136/heartjnl-2021-319844>.
17. R. D. Brook, S. Rajagopalan, C. A. Pope, et al., "Particulate Matter Air Pollution and Cardiovascular Disease," *Circulation* 121, no. 21 (2010): 2331–2378, <https://doi.org/10.1161/CIR.0b013e3181d8ece1>.
18. T. F. Mebrahtu, A. W. Morgan, R. M. West, P. M. Stewart, and M. Pujades-Rodriguez, "Oral Glucocorticoids and Incidence of Hypertension in People With Chronic Inflammatory Diseases: A Population-Based Cohort Study," *CMAJ: Canadian Medical Association Journal* 192, no. 12 (2020): E295–e301, <https://doi.org/10.1503/cmaj.191012>.
19. L. Fardet and B. Fève, "Systemic Glucocorticoid Therapy: A Review of Its Metabolic and Cardiovascular Adverse Events," *Drugs* 74, no. 15 (2014): 1731–1745, <https://doi.org/10.1007/s40265-014-0282-9>.
20. M. K. Ng and D. S. Celermajer, "Glucocorticoid Treatment and Cardiovascular Disease," *Heart* 90, no. 8 (2004): 829–830, <https://doi.org/10.1136/hrt.2003.031492>.
21. H. W. Deng, W. Y. Mei, Q. Xu, et al., "The Role of Glucocorticoids in Increasing Cardiovascular Risk," *Frontiers in Cardiovascular Medicine* 10 (2023): 1187100, <https://doi.org/10.3389/fcvm.2023.1187100>.
22. S. A. Johannesdottir, E. Horváth-Puhó, O. M. Dekkers, et al., "Use of Glucocorticoids and Risk of Venous Thromboembolism: A Nationwide Population-Based Case-Control Study," *JAMA Internal Medicine* 173, no. 9 (2013): 743–752, <https://doi.org/10.1001/jamainternmed.2013.122>.
23. K. Josey, R. Nethery, A. Visaria, et al., "Retrospective Cohort Study Investigating Synergism of Air Pollution and Corticosteroid Exposure in Promoting Cardiovascular and Thromboembolic Events in Older Adults," *BMJ Open* 13, no. 9 (2023): e072810, <https://doi.org/10.1136/bmjopen-2023-072810>.
24. Real-time Air Quality Index Visual Map, "Air Pollution in Asia," <https://aqicn.org/map/asia/>.
25. M. H. Lin, A. C. Yang, and T.-H. Wen, "Using Regional Differences and Demographic Characteristics to Evaluate the Principles of Estimation of the Residence of the Population in National Health Insurance Research Databases (NHIRD)," *Taiwan Journal of Public Health* 30 (2011): 347–360.
26. E. R. Bleeker, A. N. Menzies-Gow, D. B. Price, et al., "Systematic Literature Review of Systemic Corticosteroid Use for Asthma Management," *American Journal of Respiratory and Critical Care Medicine* 201, no. 3 (2020): 276–293, <https://doi.org/10.1164/rccm.201904-0903SO>.
27. Y. Du, X. Xu, M. Chu, Y. Guo, and J. Wang, "Air Particulate Matter and Cardiovascular Disease: The Epidemiological, Biomedical and Clinical Evidence," *Journal of Thoracic Disease* 8, no. 1 (2016): E8–e19, <https://doi.org/10.3978/j.issn.2072-1439.2015.11.37>.
28. A. J. van Zonneveld, S. A. Curriden, and D. J. Loskutoff, "Type 1 Plasminogen Activator Inhibitor Gene: Functional Analysis and Glucocorticoid Regulation of Its Promoter," *Proceedings of The National Academy of Sciences of The United States of America* 85, no. 15 (1988): 5525–5529, <https://doi.org/10.1073/pnas.85.15.5525>.
29. T. Wu, X. Yang, A. Chu, et al., "Acute Effects of Fine Particulate Matter (PM2.5) on Hospital Admissions for Cardiovascular Diseases in Lanzhou, China: A Time-Series Study," *Environmental Sciences Europe* 34 (2022): 55, <https://doi.org/10.1186/s12302-022-00634-y>.
30. Y. Tian, H. Liu, Y. Wu, et al., "Association Between Ambient Fine Particulate Pollution and Hospital Admissions for Cause Specific Cardiovascular Disease: Time Series Study in 184 Major Chinese Cities," *BMJ* 367 (2019): l6572, <https://doi.org/10.1136/bmj.l6572>.
31. B. I. Wallace, H. J. Tsai, P. Lin, et al., "Prevalence and Prescribing Patterns of Oral Corticosteroids in the United States, Taiwan, and Denmark, 2009–2018," *Clinical and Translational Science* 16, no. 12 (2023): 2565–2576, <https://doi.org/10.1111/cts.13649>.
32. National Statistics Republic of China (Taiwan), "Report on the Survey of Family Income and Expenditure," <https://eng.stat.gov.tw/News.aspx?n=4218&sms=11711>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supplement Table 1** ICD-9, ICD-10, and CPT Codes for Cohort-Defining Conditions. **Supplement Table 2** ICD-9, ICD-10, and CPT Codes for Cardiovascular thromboembolic events (CTEs). **Supplement Table 3** ICD-9 and ICD-10 codes for Comorbidities. **Supplement Table 4** Comorbidities used for adjustment of the IPWs. **Supplement Figure 1** The algorithm of residence estimation in National Health Insurance Database. **Supplement Figure 2** Example of five participants' survival paths over a one-year period. This plot demonstrates how we construct individual-level quarters over which we model corticosteroid therapy. Participants 1, 2, and 4 never receive corticosteroid therapy and thus their quarters align with seasons. An arrow denotes censoring (as in Participants 1 and 4) while an X denotes when an event occurs (as in Participant 2). Participants 3 (pink) and 5 (green) both received corticosteroids, indicated by the respective dots. Note that the quarters are evenly spaced and shifted according to the starting treatment date. The one exception is within the treatment path of Participant 3—Quarter 1 is shortened by the start of the study period. **Supplement Figure 3** Hazard ratios estimated by MSM with restricted cubic spline Cox models to evaluate associations between $PM_{2.5}$ and cardiovascular events (reference point $15\mu g/m^3$).